

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

AC6103: ACCOUNTING FOR DECISION MAKING AND CONTROL

Merlion Café Project Case

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EXECUTIVE SUMMARY

Merlion Café achieved sales growth across all three outlets in 2026, but profitability weakened. Combined sales increased by 5.66% to about $3.95 million, while profit declined by 17.92% to $544,056 and profit margin fell from 17.73% to 13.77%. The main chain-wide pressure came from direct costs, which increased by 21.19%, substantially faster than sales. However, outlet performance differed materially. Orchard accounted for 76.34% of the combined profit decline at CBD and Orchard. CBD recorded the lowest 2026 profit margin at 6.03%. Toa Payoh's sales and profit both increased, providing a useful internal benchmark.

CBD's deterioration reflects weak alignment between demand, paid capacity and marketing expenditure. Profit fell 37.11% despite modest sales growth, while modelled order-fulfilment utilisation was only 40.85% and demand remained concentrated around weekday office hours. Marketing expenditure increased by $16,800 even as orders declined. Management should therefore realign staffing with peak demand and require future campaigns to meet contribution-based break-even thresholds. Full weekend closure is not currently supported because the estimated utility savings are insufficient to offset contribution forgone unless additional avoidable labour savings are realised.

Orchard faces a different issue: comparatively strong observed demand but weaker contribution quality. Sales increased 5.36%, yet profit declined 32.67%, with adverse product mix reducing profit by $241,740. Management should rebuild the share of higher-contribution products through menu placement, bundling and cross-selling, while piloting selective price increases. Pricing sensitivity indicates that a $0.50 increase could generate about $54,956 in additional annual contribution, assuming a 5% volume decline. Shorter operating hours should also be tested using relevant-cost analysis rather than implemented immediately.

Overall, Merlion Café should pursue outlet-specific, contribution-led profitable growth. Merlion Café should manage direct-cost governance centrally, while product, pricing, marketing, staffing, and operating-hour decisions should reflect each outlet's demand profile. Major initiatives should be piloted and expanded only when incremental benefits exceed relevant costs. The proposed Strategy Map and Balanced Scorecard therefore emphasise profit margin, contribution per order, direct-cost control and modelled utilisation alongside revenue growth.

MAIN BODY

Overall Financial Performance

Merlion Café achieved revenue growth in 2026 for all three outlets, but this did not translate into higher profitability (Appendix 2.1). Combined sales increased by $211,552 (+5.66%) from 2025 to 2026. However, gross profit fell by $99,592 (-4.54%), while profit declined by $118,792 (-17.92%). Consequently, the combined gross margin deteriorated from 58.69% to 53.03%, while profit margin fell from 17.73% to 13.77%. This was driven primarily by cost pressure, reflected in a $303,186 (+21.19%) increase in direct costs, which grew substantially faster than sales, while operating expenses increased only $19,200 (+1.25%). Thus, the primary issue is not revenue generation, but the weakening conversion of sales into profit.

The aggregate result, however, masks substantial variations among the outlets. CBD recorded sales growth of 1.84%, while profit margin declined from 9.76% to 6.03%, making it the least profitable outlet on a margin basis. Orchard delivered sales growth of 5.36%, yet profit margin fell from 23.21% to 14.83%, indicating the largest margin deterioration among the outlets (Appendix 2.2). In contrast, Toa Payoh grew both revenue and profit. Its sales increased by 8.55%, while profit margin rose slightly from 17.41% to 17.61%.

From a management accounting performance-evaluation perspective, these results suggest management should evaluate outlet performance using profitability measures alongside revenue growth. CBD and Orchard together recorded a $142,258 profit decline, with CBD accounting for $33,656 (23.66%) and Orchard for $108,602 (76.34%). Toa Payoh generated a $23,466 profit increase, partially offsetting the losses and resulting in the chain-wide net profit decline of $118,792 (Appendix 2.1). Orchard therefore represents the largest source of absolute profit deterioration, while CBD requires scrutiny given its further profit margin decline to 6.03%. In contrast, Toa Payoh's simultaneous growth in sales and profit provides a useful internal benchmark. The following outlet-level analysis therefore examines two distinct causes of deterioration: CBD's weak demand-capacity and marketing alignment, and Orchard's deteriorating product mix and contribution quality.

CBD: Demand-Capacity and Marketing Misalignment

Despite modest sales growth of 1.84%, CBD's profit fell 37.11% to $57,047 in 2026, reducing its profit margin from 9.76% to 6.03%, the lowest among the three outlets (Appendices 2.1 &

2.2). This divergence suggests that CBD's primary issue is not product mix, but weak alignment between demand, paid operating capacity and marketing expenditure.

The profit bridge in Appendix 2.3 isolates the drivers of this deterioration. It shows that lower volume reduced profit by $15,234, while higher operating expenses reduced profit by another $16,800. Selling-price changes contributed a favourable $42,142, but this was almost entirely offset by a $42,006 adverse direct-cost-rate effect. Product-mix effects were comparatively immaterial at only +$307, indicating that CBD's deterioration was not primarily driven by sales composition.

Operational evidence further points to a demand-capacity mismatch. CBD used only 330,901 of 810,000 modelled annual order-fulfilment staff-minutes, equivalent to 40.85% utilisation, compared with 61.57% at Orchard and 58.48% at Toa Payoh (Appendix 4.4). This comparatively low utilisation is consistent with CBD's highly concentrated demand pattern, with activity peaking around weekday office hours (pre-shift, lunch break, and post-shift), while weekends account for only 4.18% of revenue (Appendices 4.6 & 5.1). The utilisation measure reflects modelled order-fulfilment workload, not total employee productivity.

CBD's annual marketing spending increased from $12,000 to $28,800, but orders fell by 425, from 54,869 in 2025 to 54,444 orders in 2026. From a controllability perspective, the incremental contribution generated by the additional $16,800 of marketing expenditure should be evaluated against its cost. At the 2026 direct contribution of $10.08 per order, the additional $16,800 spending requires about 1,667 incremental orders to break even, or 1,746 after estimated consumables (Appendix 6.2). Thus, CBD's key issue is resource alignment: operating capacity and marketing expenditure should be tied more closely to when and how profitable demand is actually generated.

Orchard: Product-Mix and Contribution Deterioration

Despite 5.36% sales growth, Orchard's profit declined 32.67% from $332,392 in 2025 to $223,790 in 2026, while profit margin declined from 23.21% to 14.83% (Appendices 2.1 & 2.2). Unlike CBD, Orchard continues to demonstrate comparatively strong customer demand, and its deterioration reflects weaker contribution quality driven by an adverse product-mix shift.

The profit bridge shows that the product-mix effect reduced profit by $241,740, while higher direct-cost rates reduced profit by an additional $78,407. Favourable volume (+$143,475) and selling-price (+$71,192) effects only partly offset these pressures (Appendix 2.4). Product mix was therefore the dominant negative driver of Orchard's profit decline. The shift is evident in Caramel Hot Chocolate, whose unit share fell from 32.03% to 9.98%; Cream Cakes, which declined from 34.96% to 22.95%; Caffe Latte/Cappuccino, from 10.94% to 1.99%; and Mocha/Macchiato, from 10.01% to 2.04%. In contrast, several products gained substantial share, including relatively lower-contribution products such as Green Tea and Cold Brew. The unit share of Green Tea increased from 0.95% to 19.01%, Non-Cream Cakes from 1.03% to 18.14%, Cold Brew from 1.04% to 8.96%, and Earl Grey/English Breakfast from 1.05% to 7.04% (Appendix 3.1 & 3.3).

Consistent with these shifts, products with declining contribution lost about $464,377, while products with positive contribution changes generated about $358,898 in gains, resulting in a net direct-contribution decline of about $105,479 (Appendix 3.2). These product-level changes incorporate volume, mix, selling-price and direct-cost-rate effects, while the profit bridge separately isolates the pure mix effect. Restoring Orchard's profitability therefore requires management to consider both product-level contribution and sales composition, rather than focusing on transaction or revenue growth alone.

This deterioration is not explained by insufficient customer traffic. In 2026, Orchard generated 55,136 orders, the highest among the three outlets, with an average order value of $27.37, 3.65 items per order, and 61.57% modelled fulfilment capacity utilisation. Its direct contribution per order was $14.70, below Toa Payoh's $17.02 despite Orchard recording more orders (Appendix 4.6). This reinforces that Orchard's issue lies in transaction profitability rather than insufficient customer traffic. Weekend revenue represented 51.55% of sales, with strong midday and afternoon weekend demand (Appendices 5.2 & 5.4). Management should therefore prioritise contribution per order rather than pursue additional traffic indiscriminately.

Cross-Outlet Operational Analysis

Taken together, the outlet diagnoses indicate a chain-wide operating-design misalignment. The largely uniform menu, pricing, opening-hour and staffing policies are being applied to

locations with materially different demand and profitability profiles. CBD primarily serves concentrated weekday office demand, with weekends contributing only 4.18% of sales. Orchard experiences strong midday and weekend demand, with weekends contributing 51.55%. Toa Payoh has broader daytime and residential demand, with a 46.78% weekend revenue share (Appendix 5.3). Toa Payoh's 9.79% profit growth suggests that the existing model remains comparatively well aligned with its market, making it a useful internal benchmark, although its different demand profile limits direct transferability.

Alongside these location-specific issues, direct-cost pressure is common across the chain. Combined direct costs rose 21.19%, far faster than sales, which increased 5.66%, while gross margin declined at all three outlets (Appendix 2.1). Hence, cost control should be addressed centrally, while product, pricing, marketing and capacity decisions remain outlet-specific.

Service mode also affects modelled resource intensity. Dine-in fulfilment requires three minutes per order plus two minutes per item, while takeaway and delivery require approximately 60% less modelled fulfilment time. Consequently, direct contribution per modelled service minute ranges from $2.87 to $3.85 for takeaway and delivery, compared with $1.17 to $1.54 for dine-in (Appendix 4.2). However, this does not prove takeaway or delivery is more profitable, as the model omits delivery-platform commissions and several support activities. Management should therefore use contribution relative to fulfilment resources as an operational efficiency measure rather than a definitive measure of channel profitability.

Overall, the analysis identifies one chain-wide and two outlet-specific challenges. Direct-cost pressure affects all three outlets. CBD needs better alignment between capacity and marketing spend to match concentrated office-hour demand. Orchard needs stronger product-mix and contribution management despite healthy customer traffic. Toa Payoh provides the strongest internal benchmark, although its residential demand profile limits direct transferability. Merlion Café should therefore combine central cost discipline and common performance measures with outlet-specific product, pricing, marketing and operating decisions. Performance should be evaluated using profit margin, contribution per order, contribution by hour and modelled utilisation alongside revenue growth.

RECOMMENDATIONS

Recommendations for CBD Outlet

CBD should adopt demand-based capacity and marketing management to align resources with concentrated weekday office demand. Management should first adjust shift timing and coverage around observed peak periods rather than maintain a uniform staffing pattern. Given modelled order-fulfilment utilisation of only 40.85%, the priority is to redeploy existing paid capacity before pursuing additional staffing or demand-generation expenditure.

Marketing should also become contribution-accountable. Annual marketing spending increased by $16,800 while order volume fell by 425. Future campaigns should target CBD's weekday office-worker segment and be approved against contribution-based performance measures. At the 2026 direct contribution of $10.08 per order, the additional expenditure requires approximately 1,667 incremental orders to break even, or 1,746 after estimated consumables (Appendix 6.2). Campaigns that do not demonstrate sufficient incremental contribution should be redesigned or discontinued.

Operating-hour reductions should be considered after staffing and marketing alignment. Appendix 6.3 shows that full weekend closure would put approximately $22,951 of contribution at risk while saving only $12,000 in utilities, resulting in an estimated $10,951 loss before any labour savings. Weekend closure is therefore not financially supported under the current assumptions unless additional avoidable labour savings can be demonstrated.

Recommendations for Orchard Outlet

Orchard should adopt a contribution-led product and pricing strategy rather than pursue sales growth alone. Management should selectively rebuild the share of higher-contribution products, including Caramel Hot Chocolate, Mocha/Macchiato, and Caffe Latte/Cappuccino, through menu placement, bundling, and cross-selling (Appendix 3.1 & 3.3). Performance should be assessed by whether these initiatives raise direct contribution per order above the current $14.70 level without materially weakening demand (Appendix 4.6).

Additionally, selective price increases should be piloted rather than implemented uniformly. Appendix 6.1 estimates that a $0.50 price increase could generate about $54,956 of additional annual contribution under an assumed 5% volume decline, while a $0.75 increase could

generate about $102,692 under the same assumed volume response. As these are sensitivity scenarios rather than forecasts, management should monitor actual volume, unit mix and contribution responses and refine product-level pricing based on observed results.

Furthermore, operating hours should similarly be evaluated using relevant-cost analysis. Appendix 6.3 estimates that an 11 am to 8 pm schedule could generate an annual net benefit of $12,903 to $13,812 before labour savings, depending on whether consumables are avoidable. Because customer migration to other periods is not modelled, shorter hours should be adopted only if a pilot confirms that avoidable cost savings exceed contribution forgone without materially disrupting peak-period demand.

Recommendations for Chain-Wide Strategy

Merlion Café should establish central direct-cost governance while retaining outlet-specific commercial decisions. Direct costs increased by 21.19% in 2026 compared with only 5.66% sales growth, contributing to gross-margin deterioration across all three outlets. Management should therefore monitor direct costs as a percentage of sales and contribution by product monthly, with material adverse variances triggering review of supplier prices, ingredient specifications, portion control and waste. Management should initially investigate these as potential drivers rather than assumed causes. At 2026 sales of approximately $3.95 million, every one-percentage-point reduction in the direct-cost-to-sales ratio would improve annual gross profit by approximately $39,500, holding other variables constant.

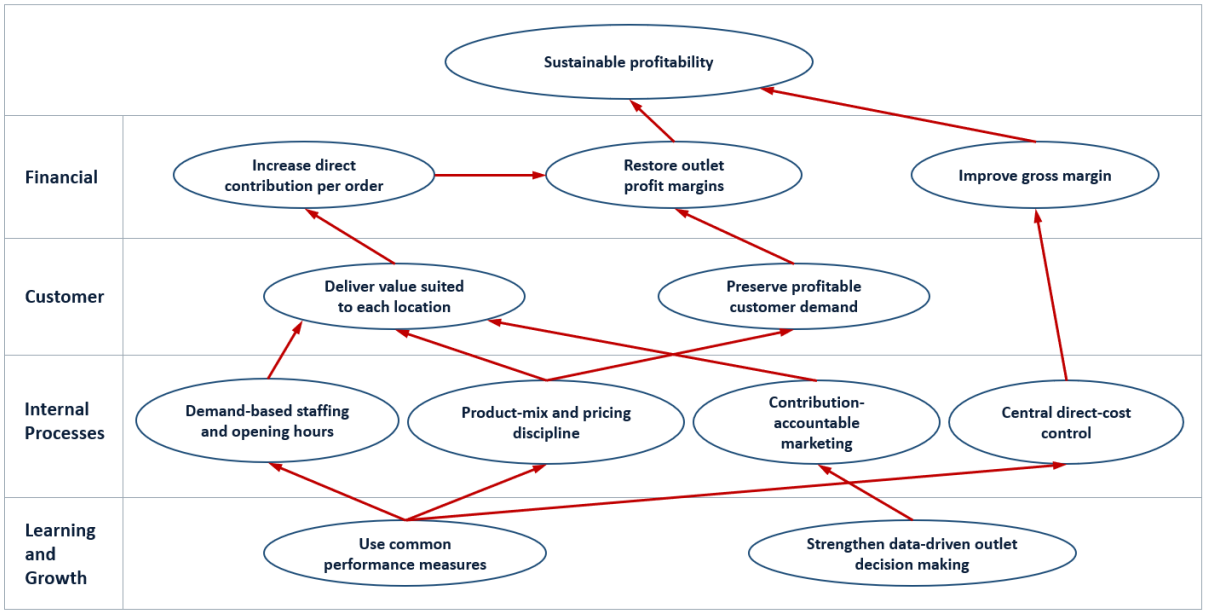
Commercial and operating decisions should therefore be differentiated by outlet, reflecting different hourly contribution patterns (Appendix 5.5). CBD should optimise capacity and marketing around office-hour demand. Orchard should prioritise contribution per order and product mix. Toa Payoh should retain its core model while serving as an internal benchmark for basket economics and capacity use. Benchmarking should be selective rather than mechanical because each outlet serves a different demand profile.

Service mode efficiency should also be monitored using contribution per modelled service minute. Takeaway and delivery require substantially less modelled fulfilment time than dine-in, so this measure can support staffing and process decisions. However, management should not deliberately shift customers toward delivery solely on this basis because platform

commissions and several support activities are unavailable. The metric should be used as an operational efficiency indicator rather than a complete channel profitability measure.

Major pricing, operating-hour, staffing and marketing changes should follow a pilot-and-review process with predefined success measures. Decisions should be assessed using incremental contribution, profit margin, contribution per order, contribution by hour, demand retention and modelled utilisation rather than revenue alone. Initiatives should be expanded only where realised incremental benefits exceed relevant costs.

Strategy Map and Balanced Scorecard



Perspective	Strategic Objectives	Measures	Targets	Initiatives
Financial	Restore profitability	Profit margin	Improve from 13.77% year-on-year	Contribution-led outlet management
	Improve direct contribution per order	Direct contribution / order	Orchard: > $14.70 without material demand loss	Selective pricing, bundling and cross-selling
	Improve gross-margin quality	Direct cost / sales	Reduce from 43.89%; 1pp reduction ≈ +$39,500 gross profit	Central direct-cost discipline
Customer	Preserve profitable customer demand	Order volume / demand retention during pilots	No material decline in profitable demand	Monitor actual demand during pricing, hours and staffing pilots
	Deliver value suited to each location	Contribution/order + outlet demand pattern	Improve contribution while maintaining local demand fit	CBD office-worker focus; Orchard mix strategy; TPY selective benchmark
Internal Processes	Improve product mix	Higher-contribution products' sales share	Increase share of selected higher-contribution products	Bundling, cross-selling and menu placement
	Match staffing and hours to demand	Peak-period coverage / modelled utilisation	Improve CBD utilisation from 40.85% without constraining peaks	Demand-based scheduling
	Make marketing contribution-accountable	Incremental contribution / campaign ROI	Contribution ≥ campaign cost; CBD ≈1,667 incremental orders	Contribution-based campaign approval
	Improve operating-hour economics	Avoidable savings vs contribution forgone	Avoidable savings > contribution forgone	Pilot shorter hours and verify avoidable savings
Learning & Growth	Strengthen evidence-based outlet decision making	Dashboard review / % material changes piloted	Monthly outlet review; 100% major changes piloted	Common dashboard + pilot-and-review framework

APPENDICES

1. Appendix 1: Data Reconciliation & Methodology

1.1. Data Reconciliation

Outlet	Year	Case Sales	Python Sales	Diff.	Direct Cost Case	Python Direct Cost	Diff.
CBD	2025	929,424	929,423.80	-0.2	365,862	365,862.10	0.1
ORD	2025	1,432,103	1,432,102.90	-0.1	516,307	516,306.80	-0.2
TPY	2025	1,376,756	1,376,756.00	0	548,403	548,403.30	0.3
CBD	2026	946,504	946,504.00	0	397,734	397,734.03	0.03
ORD	2026	1,508,913	1,508,913.20	0.2	698,597	698,597.19	0.19
TPY	2026	1,494,418	1,494,417.60	-0.4	637,427	637,427.33	0.33

Note: Transaction-level sales and direct costs were independently reconstructed using quantities and the annual price/direct-cost tables. Python-calculated totals reconcile to the case financial statements within rounding differences.

1.2. Key Analytical Assumptions

Assumption	**Treatment**
Unique order definition	Outlet + Date + Order ID
Eat In service time	3 minutes + 2 minutes per item in order
Takeaway/Delivery service time	40% of Eat In-equivalent time
Practical capacity	67,500 staff-minutes/month/outlet 810,000 staff-minutes/year/outlet
Capacity cost rate	$0.2222/minute
Capacity interpretation	Order-fulfilment workload only; not total employee utilisation
Customer analysis limitation	No customer ID; Service mode is treated as a customer-service segment
Operating-hours scenarios	Sensitivity analyses; no automatic labour saving is assumed but utility savings estimated

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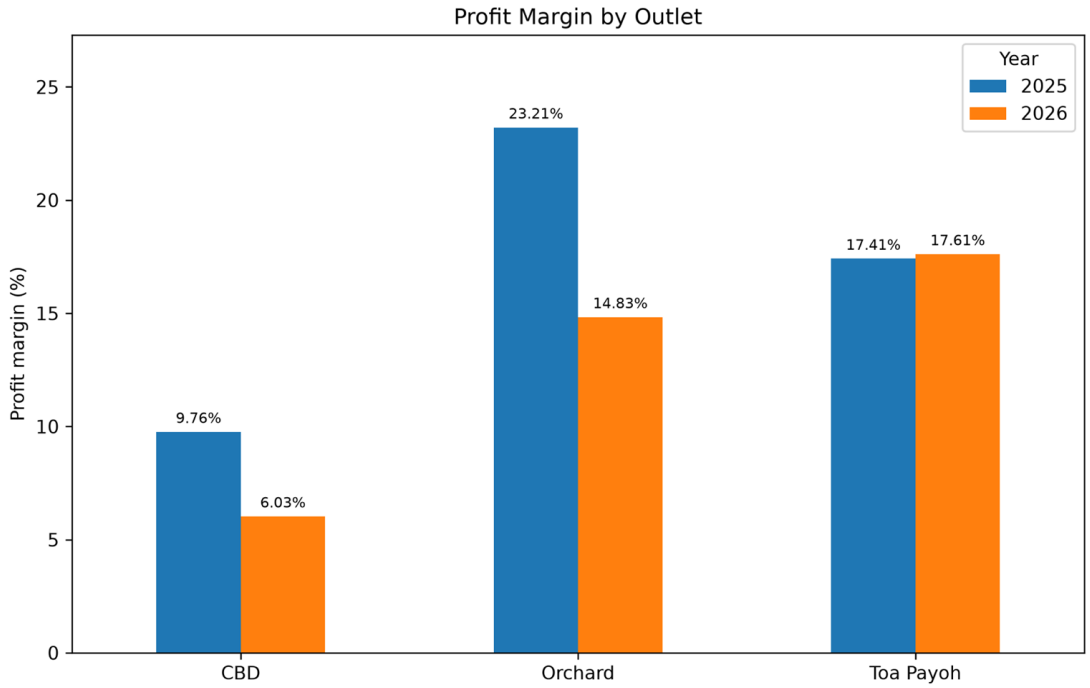
2. Appendix 2: Financial Performance & Profit Bridge

2.1. Financial Performance Comparison

	CBD		ORD		TPY	
Year	**2025**	**2026**	**2025**	**2026**	**2025**	**2026**
Sales	929,424	946,504	1,432,103	1,508,913	1,376,756	1,494,418
Direct cost	365,862	397,734	516,307	698,597	548,403	637,427
Consumables	22,859	24,923	44,604	47,726	46,200	48,972
Gross profit	540,703	523,847	871,192	762,590	782,153	808,019
Operating expense	450,000	466,800	538,800	538,800	542,400	544,800
Profit	90,703	57,047	332,392	223,790	239,753	263,219
Gross margin	58.18%	55.35%	60.83%	50.54%	56.81%	54.07%
Profit margin	9.76%	6.03%	23.21%	14.83%	17.41%	17.61%
Sales yoy change		1.84%		5.36%		8.55%
Gross profit yoy change		-3.12%		-12.47%		3.31%
Profit yoy change		-37.11%		-32.67%		9.79%
Profit change		-33,656		-108,602		23,466
Chain-wide Analysis						
		2025	2026	Change	% Change	
Sales		3,738,283	3,949,835	+211,552	+5.66%	
Direct Costs		1,430,572	1,733,758	+303,186	+21.19%	
Gross Profit		2,194,048	2,094,456	-99,592	-4.54%	
Operating Expenses		1,531,200	1,550,400	+19,200	+1.25%	
Profit		662,848	544,056	-118,792	-17.92%	
Gross Margin		58.69%	53.03%	-5.66 pp		
Profit Margin		17.73%	13.77%	-3.96 pp		

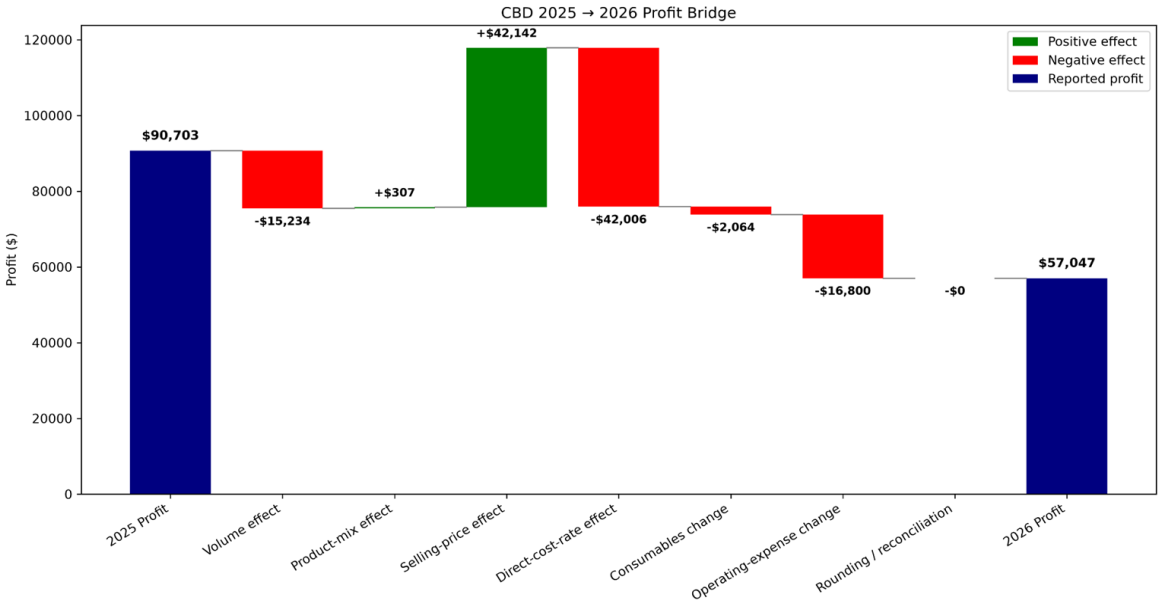
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2.2. Profit Margin by Outlet 2025-2026



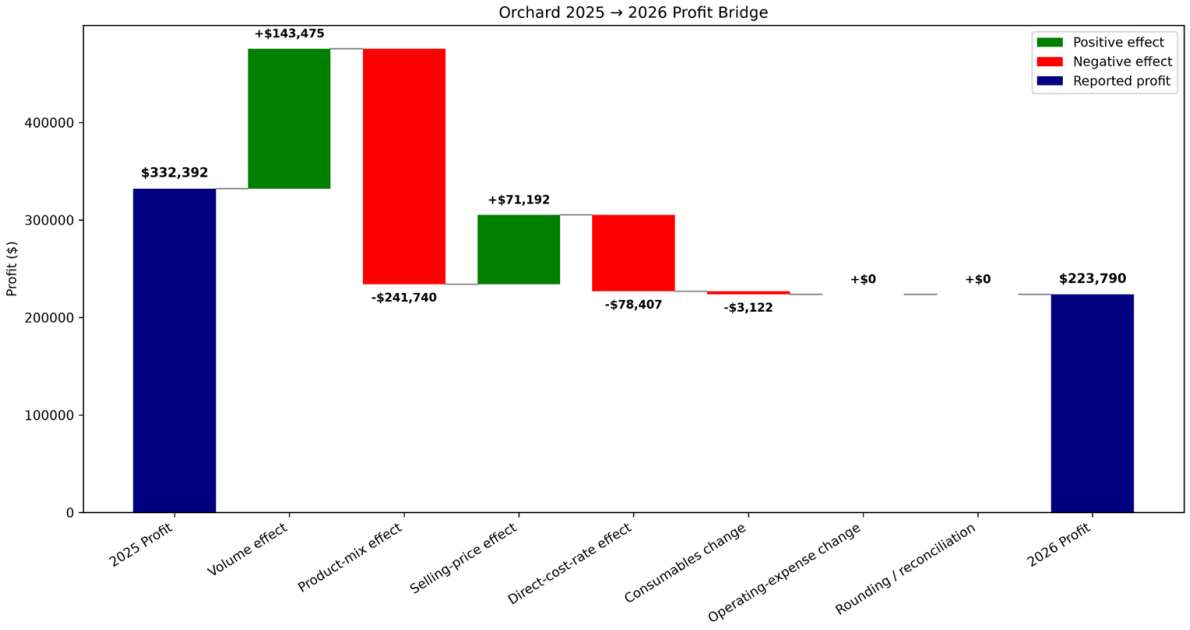
CBD's profit margin declined from about 9.8% to 6.0%, while Orchard deteriorated more sharply, from about 23.2% to 14.8%. TPY remained stable at approximately 17–18%.

2.3. CBD Profit Bridge



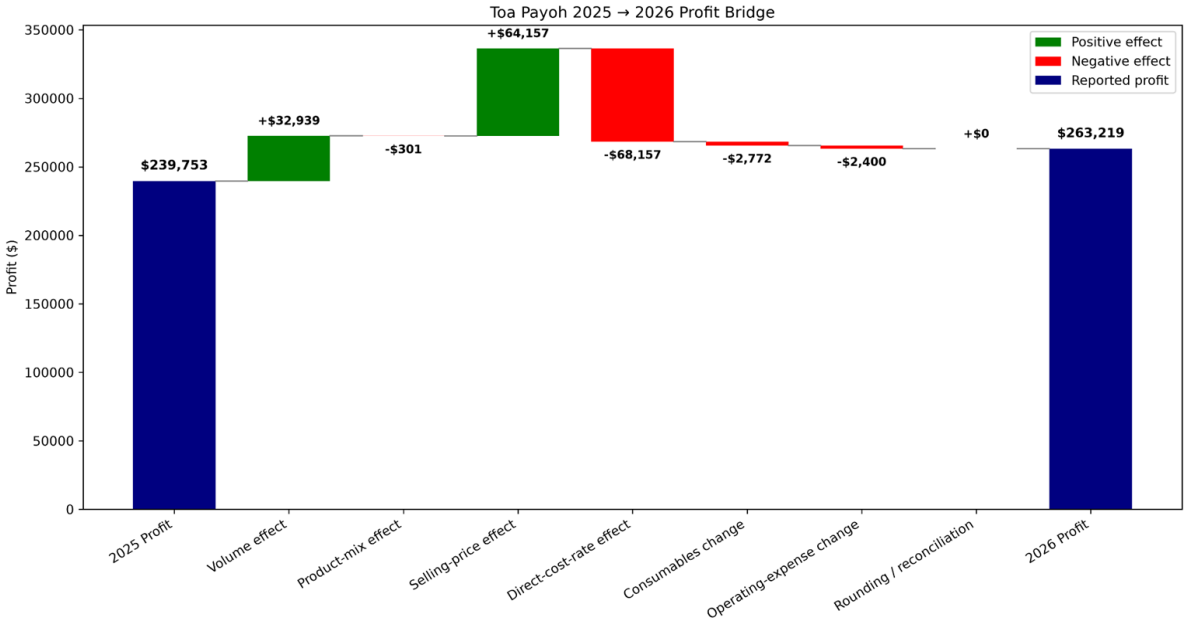
CBD's decline was driven mainly by weaker volume and higher operating expenses. Selling-price increases had a favourable effect, but higher direct-cost rates largely absorbed it. Product-mix effects were comparatively small.

2.4. Orchard Profit Bridge



Orchard's adverse product-mix effect was the dominant negative driver and more than offset the favourable effect of higher volume. Pricing contributed positively, but higher direct-cost rates further weakened profitability.

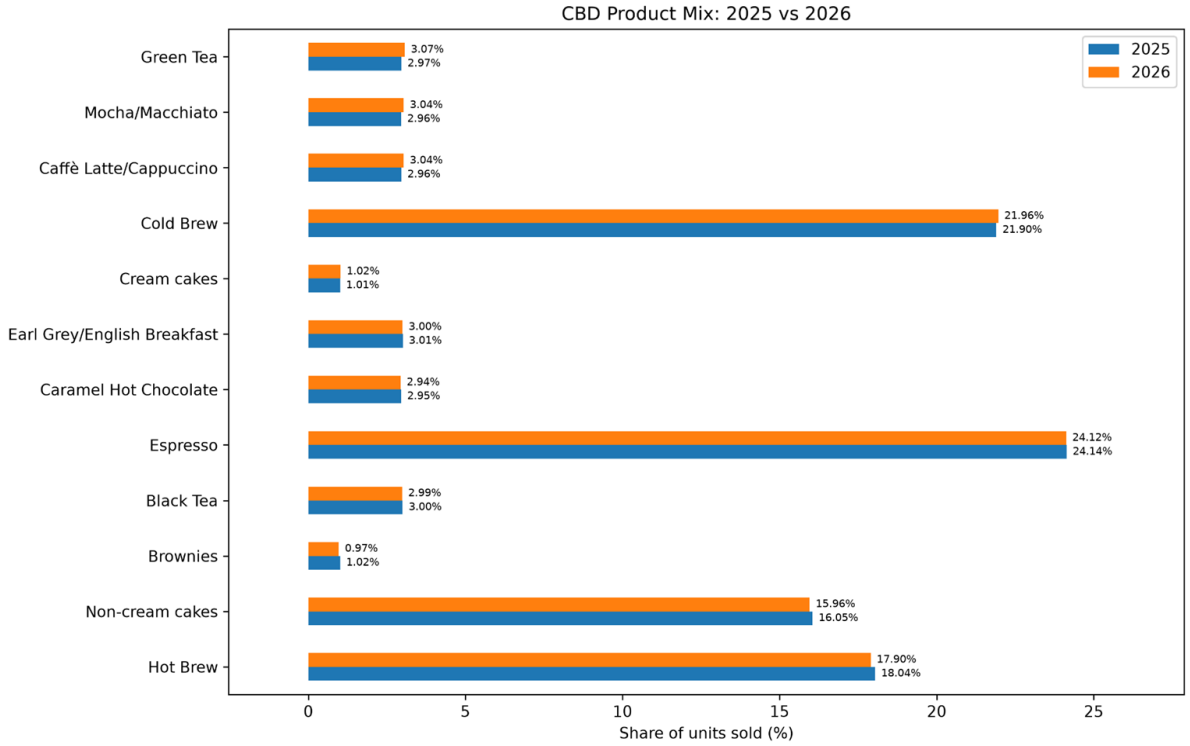
2.5. Toa Payoh Profit Bridge

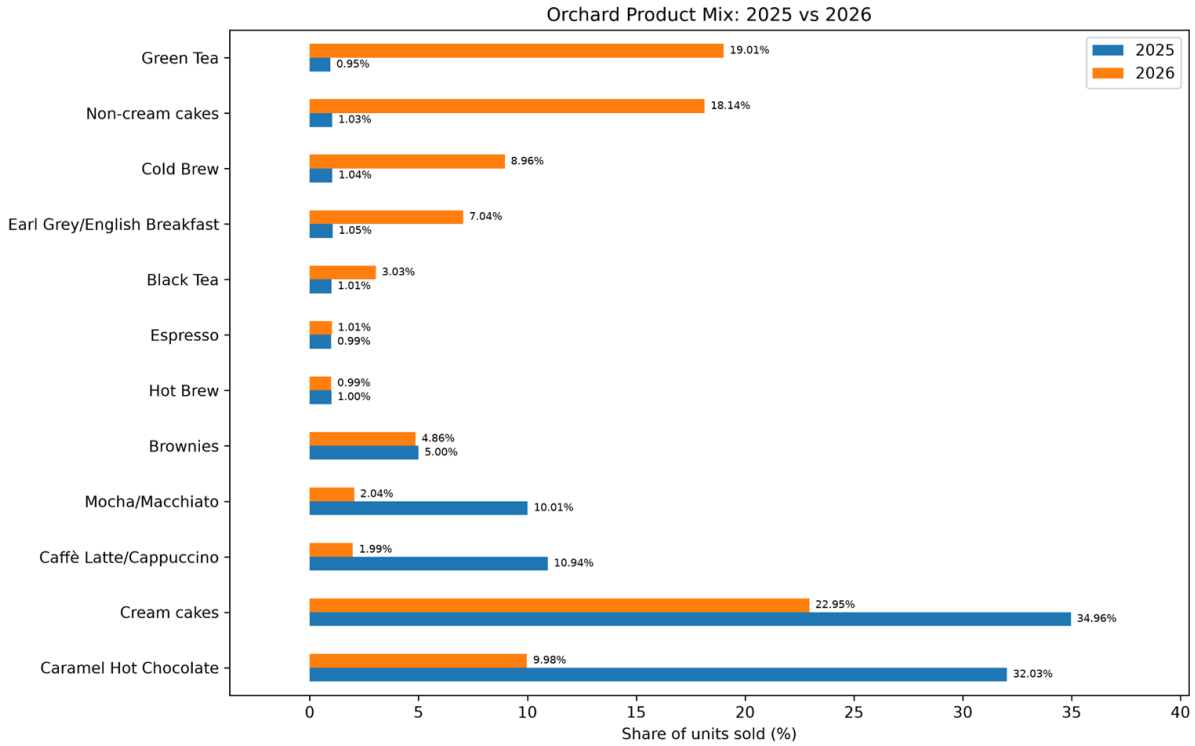


TPY benefited from favourable volume growth and pricing, which were sufficient to absorb increases in direct-cost rates and operating costs, resulting in modest profit growth.

3. Appendix 3: Product Mix & Product Profitability Analysis

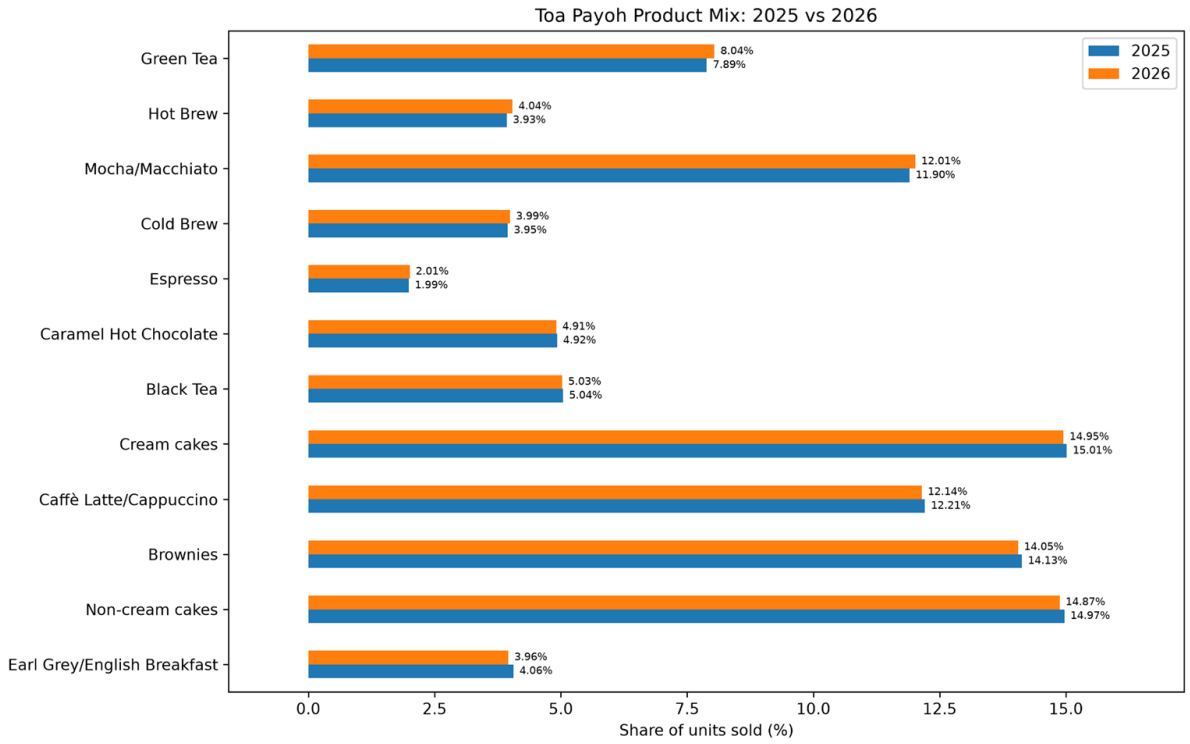
3.1. Product Mix 2025 vs 2026



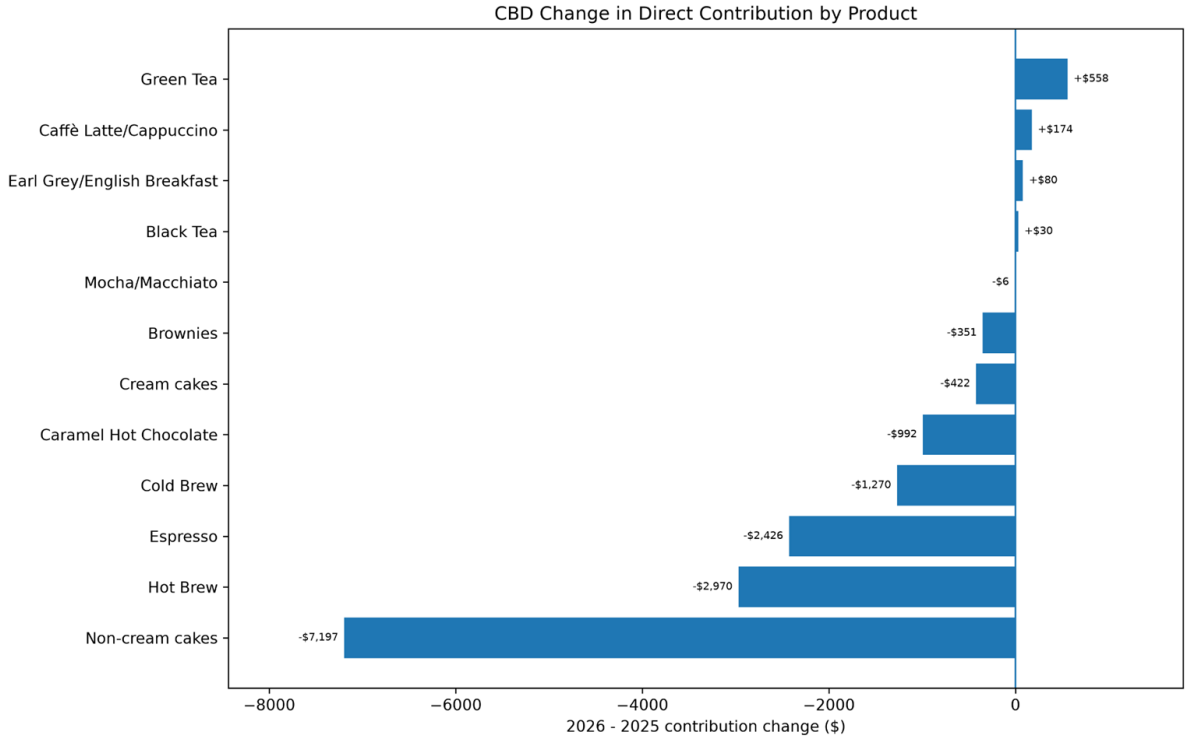


Orchard experienced a substantial shift away from Caramel Hot Chocolate, Cream Cakes, Latte/Cappuccino and Mocha/Macchiato, while Green Tea, Non-Cream Cakes and Cold Brew gained substantial unit share.

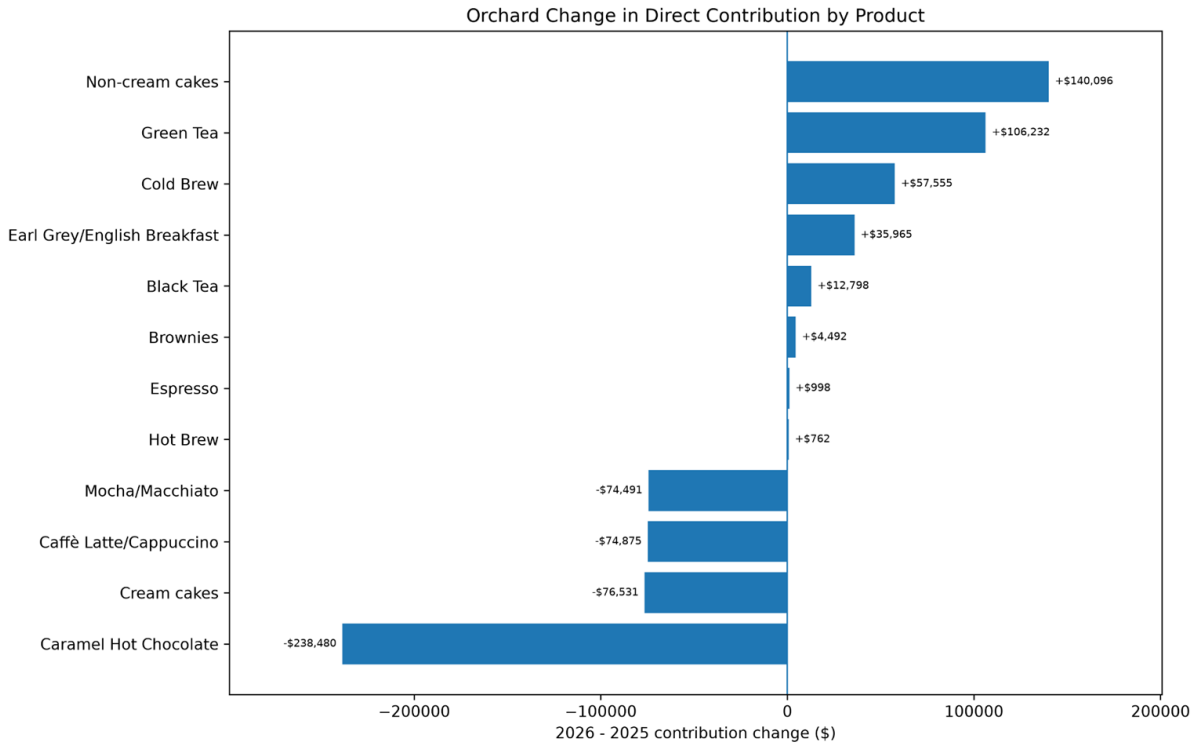
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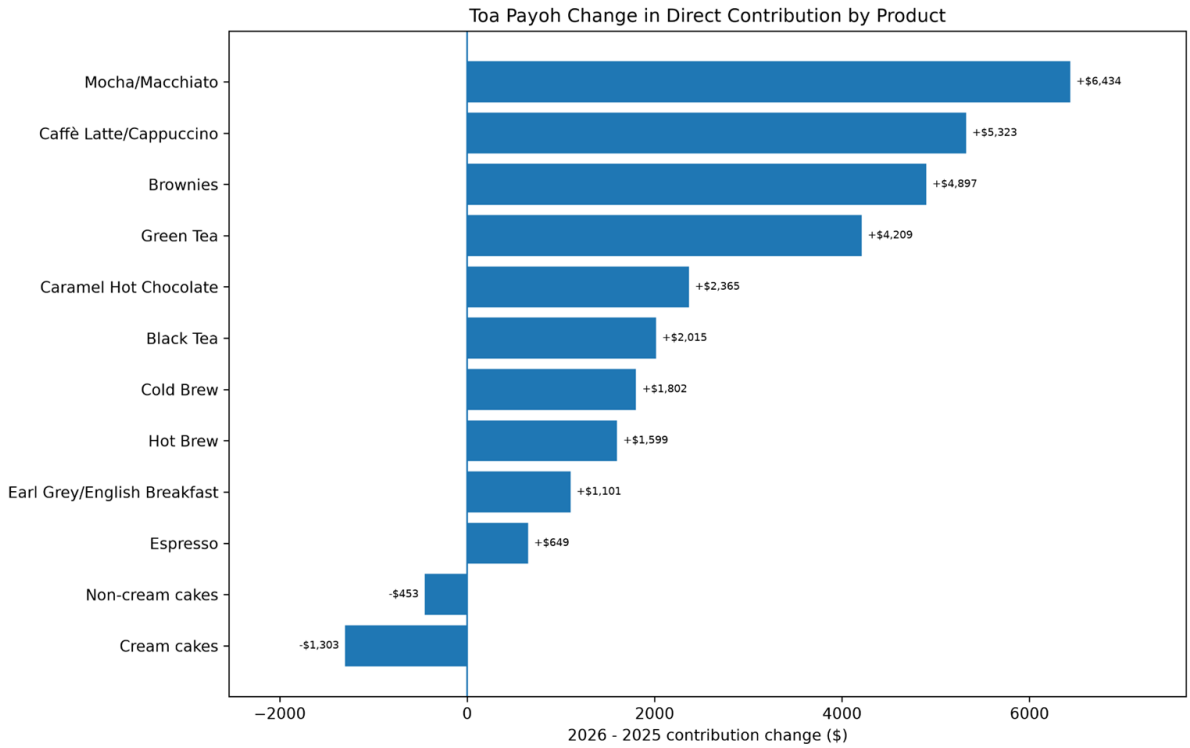
3.2. Product Contribution Change



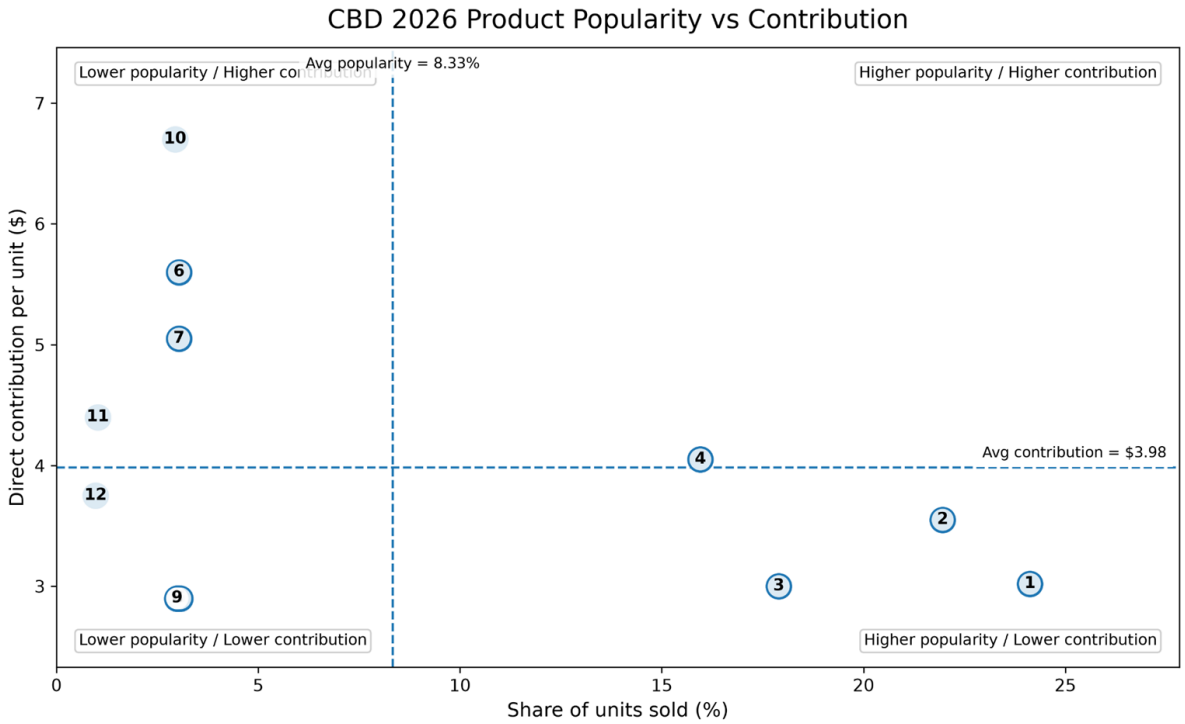
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The Orchard contribution analysis shows that gains from Green Tea, Non-Cream Cakes, and Cold Brew only partly offset declines in Caramel Hot Chocolate, Cream Cakes, and selected coffee products.

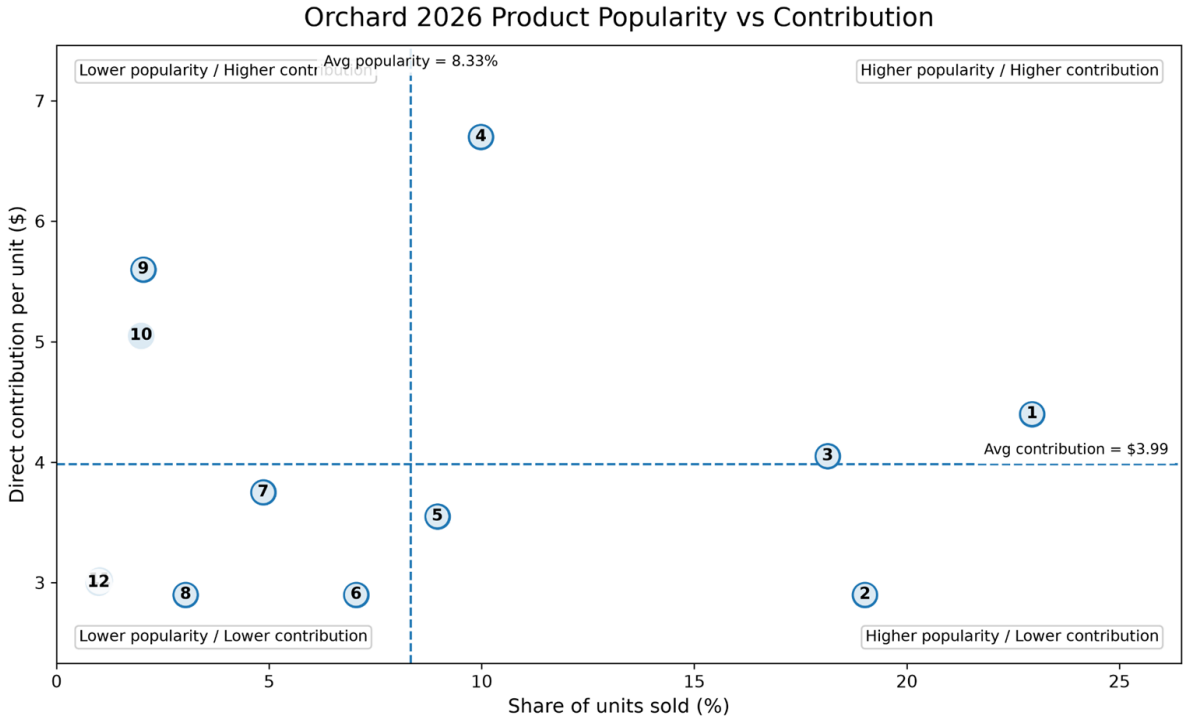


3.3. Product Popularity vs Contribution Analysis



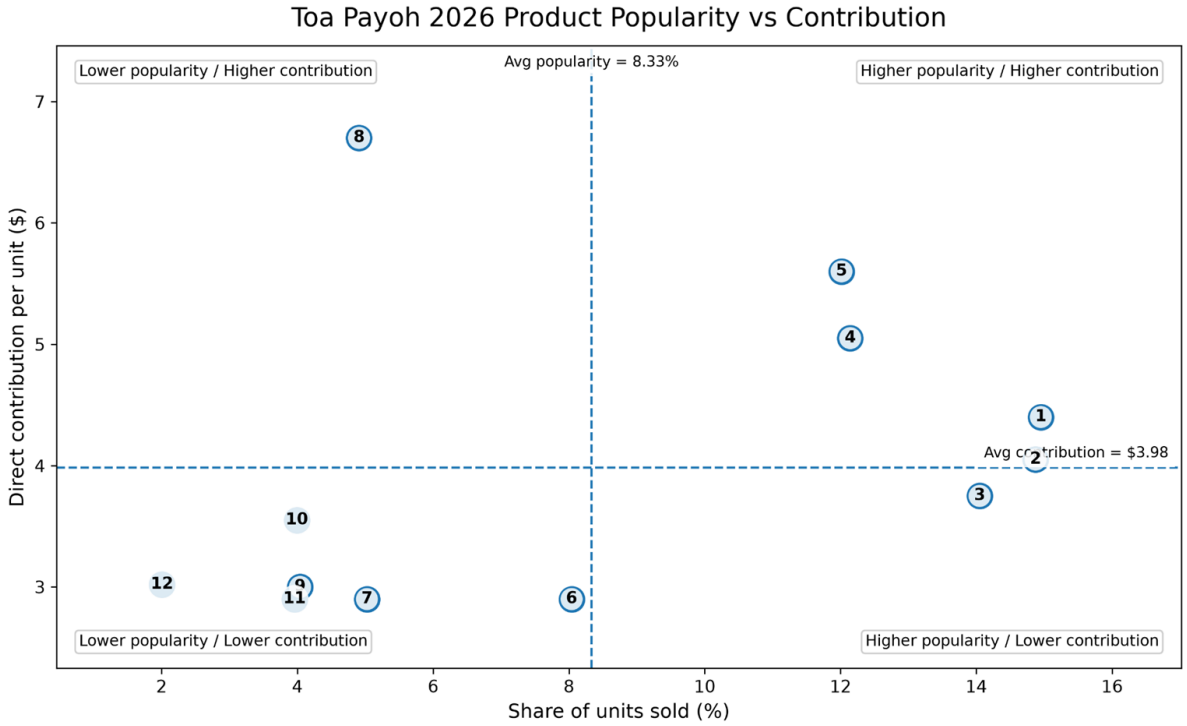
ID	Product	Popularity (% of units)	Contribution / unit
1	Espresso	24.12%	$3.02
2	Cold Brew	21.96%	$3.55
3	Hot Brew	17.90%	$3.00
4	Non-cream cakes	15.96%	$4.05
5	Green Tea	3.07%	$2.90
6	Mocha/Macchiato	3.04%	$5.60
7	Caffè Latte/Cappuccino	3.04%	$5.05
8	Earl Grey/English Breakfast	3.00%	$2.90
9	Black Tea	2.99%	$2.90
10	Caramel Hot Chocolate	2.94%	$6.70
11	Cream cakes	1.02%	$4.40
12	Brownies	0.97%	$3.75

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ID	Product	Popularity (% of units)	Contribution / unit
1	Cream cakes	22.95%	$4.40
2	Green Tea	19.01%	$2.90
3	Non-cream cakes	18.14%	$4.05
4	Caramel Hot Chocolate	9.98%	$6.70
5	Cold Brew	8.96%	$3.55
6	Earl Grey/English Breakfast	7.04%	$2.90
7	Brownies	4.86%	$3.75
8	Black Tea	3.03%	$2.90
9	Mocha/Macchiato	2.04%	$5.60
10	Caffè Latte/Cappuccino	1.99%	$5.05
11	Espresso	1.01%	$3.02
12	Hot Brew	0.99%	$3.00

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ID	Product	Popularity (% of units)	Contribution / unit
1	Cream cakes	14.95%	$4.40
2	Non-cream cakes	14.87%	$4.05
3	Brownies	14.05%	$3.75
4	Caffè Latte/Cappuccino	12.14%	$5.05
5	Mocha/Macchiato	12.01%	$5.60
6	Green Tea	8.04%	$2.90
7	Black Tea	5.03%	$2.90
8	Caramel Hot Chocolate	4.91%	$6.70
9	Hot Brew	4.04%	$3.00
10	Cold Brew	3.99%	$3.55
11	Earl Grey/English Breakfast	3.96%	$2.90
12	Espresso	2.01%	$3.02

Products in the upper-right region combine strong demand with strong unit contribution, while products in the lower-left region have relatively weak demand and contribution.

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3.4. Selected Orchard Product Table

Product	2025 Mix	2026 Mix	Mix Change (percentage points)	Contribution /Unit 2026	Contribution Change ($), 2025-2026
Caramel Hot Chocolate	32.03%	9.98%	-22.05	6.7	-238,479.80
Cream Cakes	34.96%	22.95%	-12.02	4.4	-76,531.00
Latte/Cappuccino	10.94%	1.99%	-8.95	5.05	-74,875.35
Green Tea	0.95%	19.01%	18.07	2.9	106,231.90
Non-Cream Cakes	1.03%	18.14%	17.10	4.05	140,095.80
Cold Brew	1.04%	8.96%	7.91	3.55	57,554.55

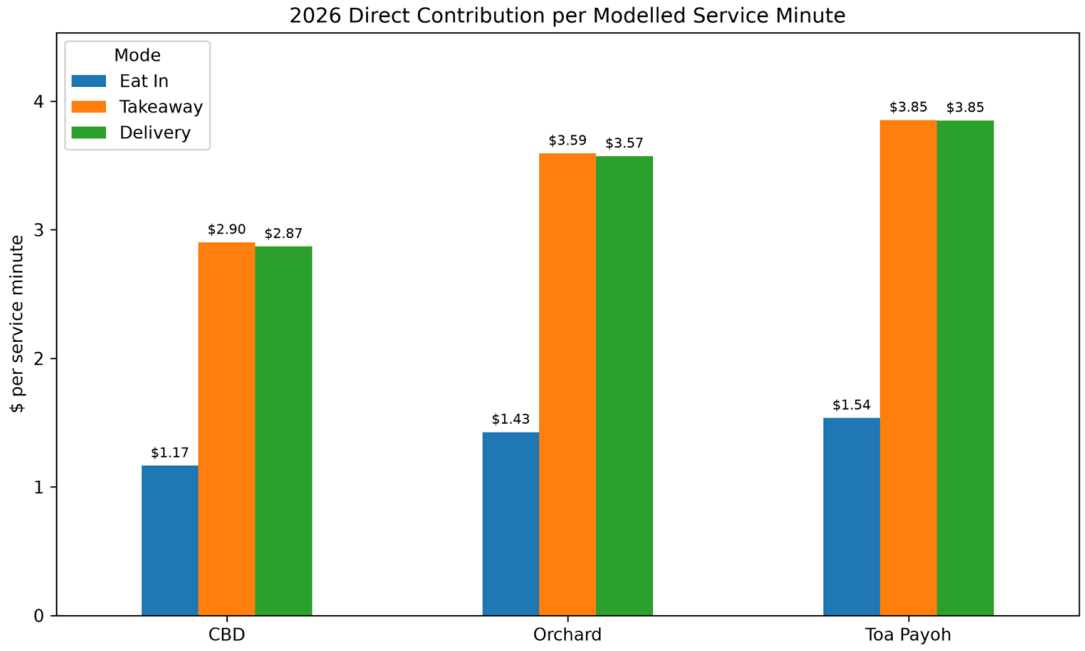
4. Appendix 4: Customer/Channel & Capacity Analysis

4.1. Channel Economics Table

Outlet	Year	Mode	Orders	Avg items/ order	Avg order value	Contribution/ order	Avg service min/ order	Contribution/ service min
CBD	2025	Delivery	2,761	2.92	17.09	10.36	3.53	2.93
CBD	2025	Eat In	27,323	2.91	17.03	10.33	8.82	1.17
CBD	2025	Takeaway	24,785	2.87	16.82	10.19	3.49	2.92
CBD	2026	Delivery	2,761	2.75	16.84	9.77	3.40	2.87
CBD	2026	Eat In	27,261	2.84	17.46	10.12	8.69	1.17
CBD	2026	Takeaway	24,422	2.83	17.36	10.06	3.47	2.90
ORD	2025	Delivery	5,224	3.32	27.38	17.51	3.85	4.54
ORD	2025	Eat In	42,235	3.29	27.14	17.35	9.59	1.81
ORD	2025	Takeaway	5,284	3.28	27.01	17.29	3.82	4.52
ORD	2026	Delivery	5,558	3.66	27.52	14.76	4.13	3.57

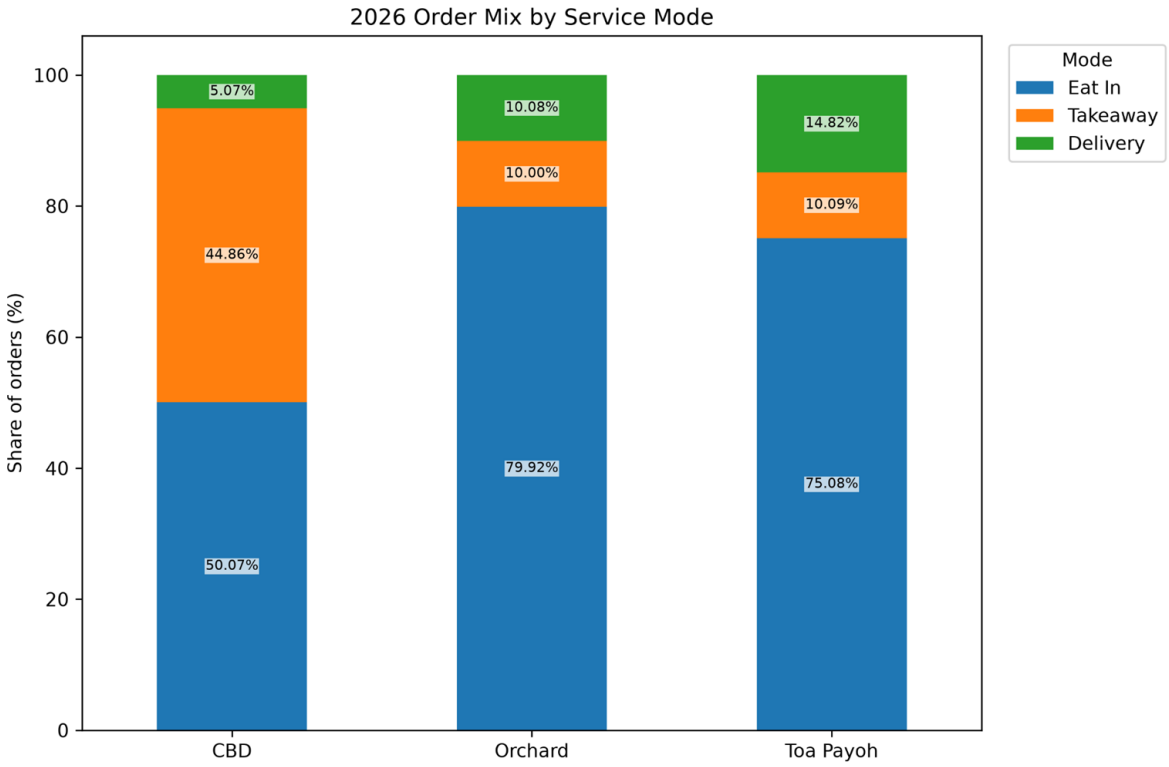
ORD	2026	Eat In	44,062	3.64	27.31	14.66	10.28	1.43
ORD	2026	Takeaway	5,516	3.68	27.70	14.89	4.14	3.59
TPY	2025	Delivery	7,475	3.87	27.34	16.44	4.30	3.82
TPY	2025	Eat In	37,384	3.91	27.58	16.59	10.83	1.53
TPY	2025	Takeaway	4,961	4.05	28.53	17.18	4.44	3.87
TPY	2026	Delivery	7,464	4.06	29.85	17.13	4.45	3.85
TPY	2026	Eat In	37,809	4.03	29.63	16.99	11.05	1.54
TPY	2026	Takeaway	5,082	4.05	29.75	17.11	4.44	3.85

4.2. Contribution per Service Minute

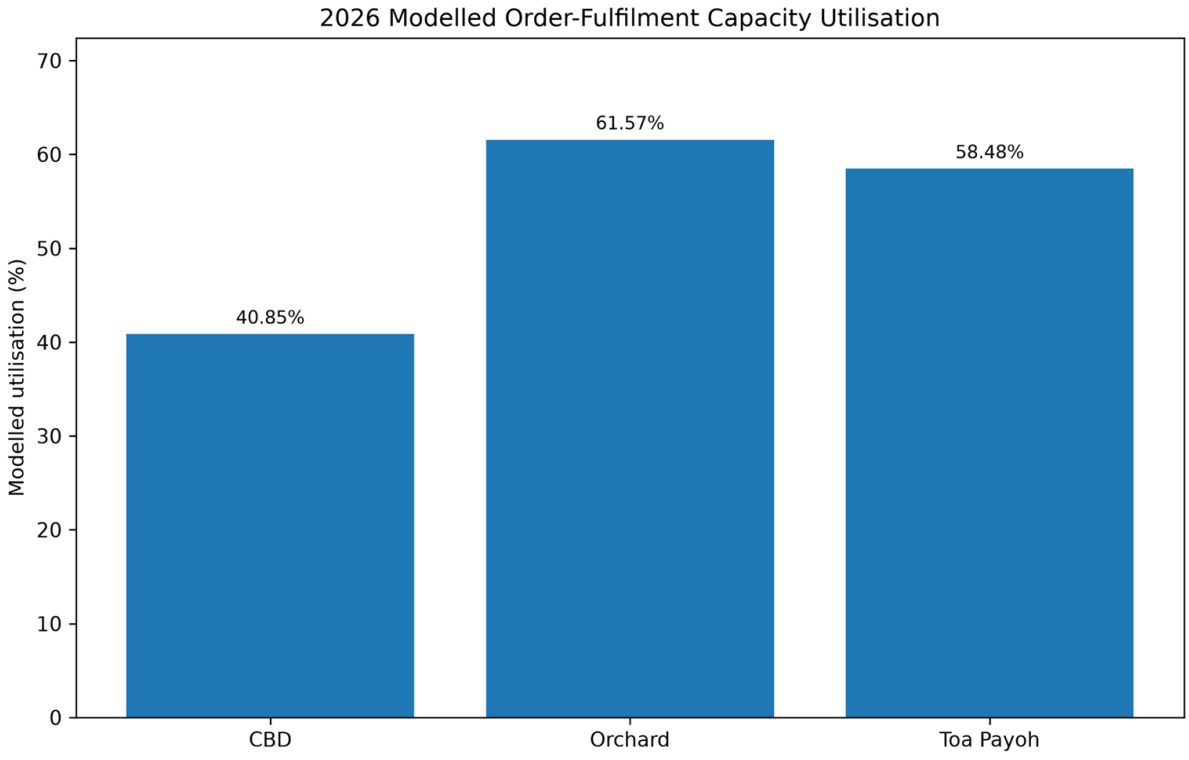


Takeaway and delivery require substantially fewer modelled fulfilment minutes than dine-in and therefore generate more direct contribution per modelled fulfilment minute. However, Delivery commissions are unavailable, so this is not complete net delivery profitability.

4.3. Channel Order Mix



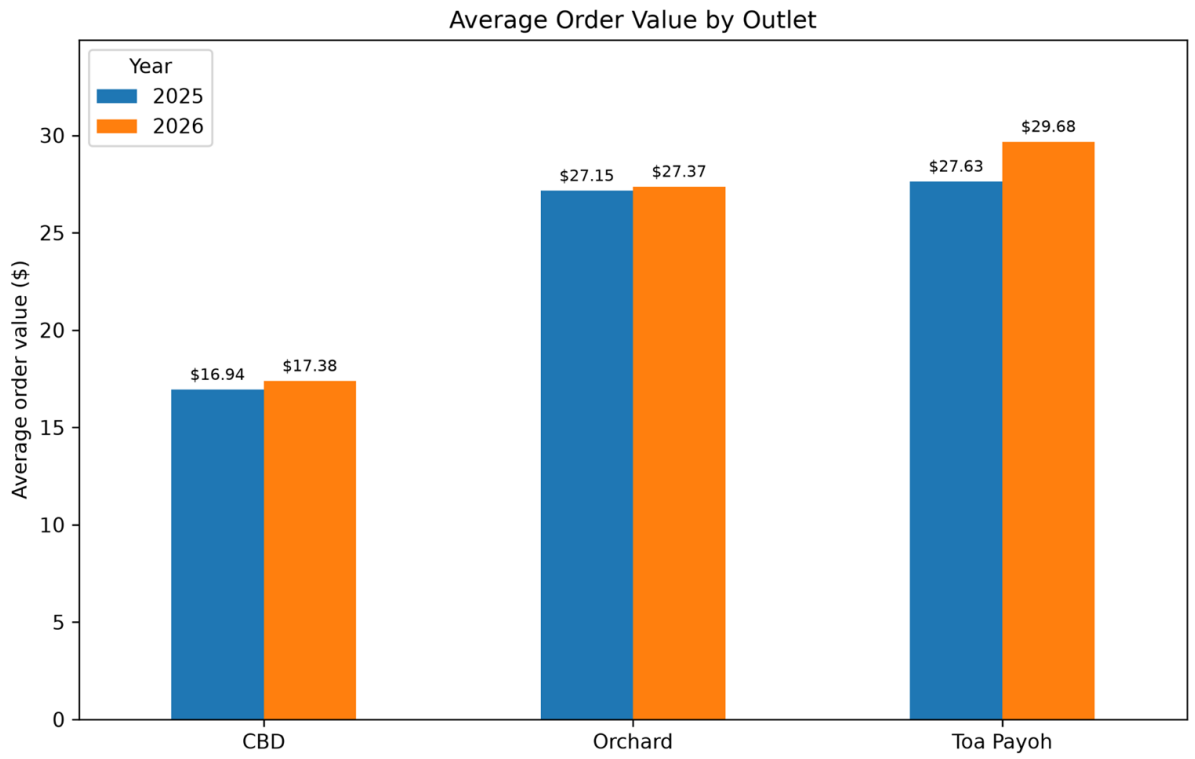
4.4. Capacity Utilisation



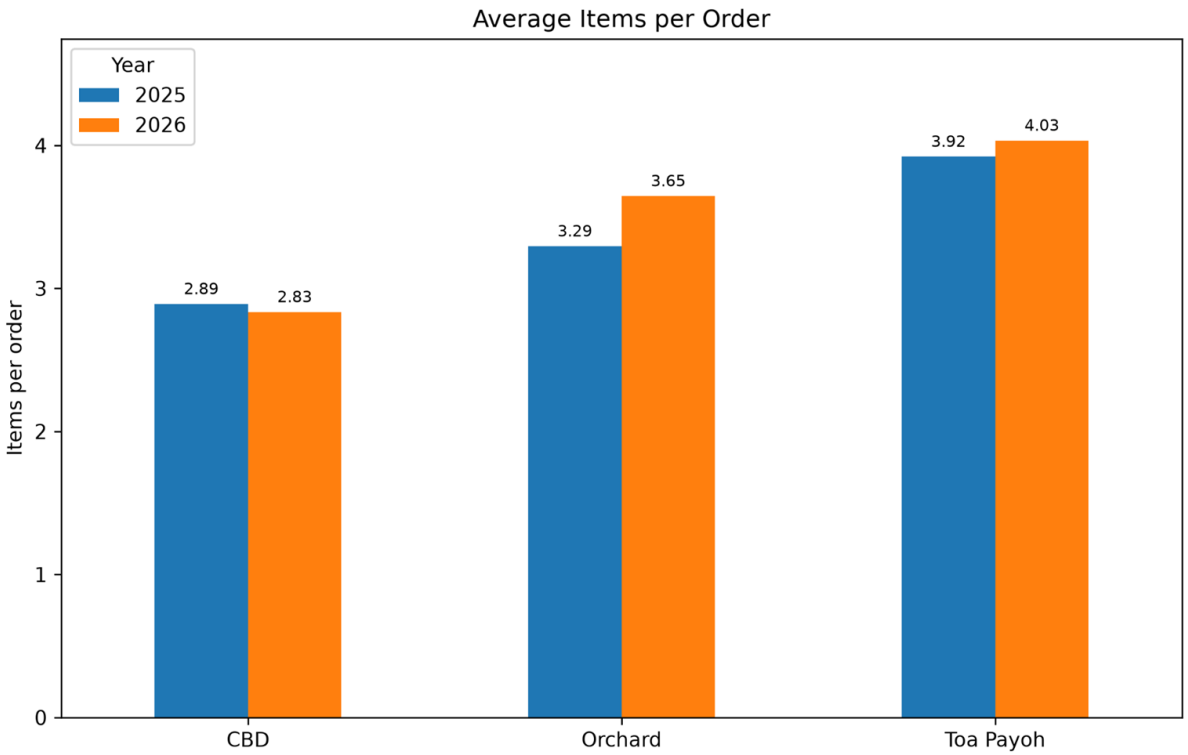
Outlet	Practical Capacity	Modelled Service Minutes	Modelled Utilisation
CBD	810,000	330,901	40.85%
ORD	810,000	498,684	61.57%
TPY	810,000	473,682.6	58.48%

CBD shows materially lower modelled order-fulfilment utilisation than Orchard and TPY. This modelled utilisation measure captures order-fulfilment workload only and should not be interpreted as total employee utilisation.

4.5. Order Economics



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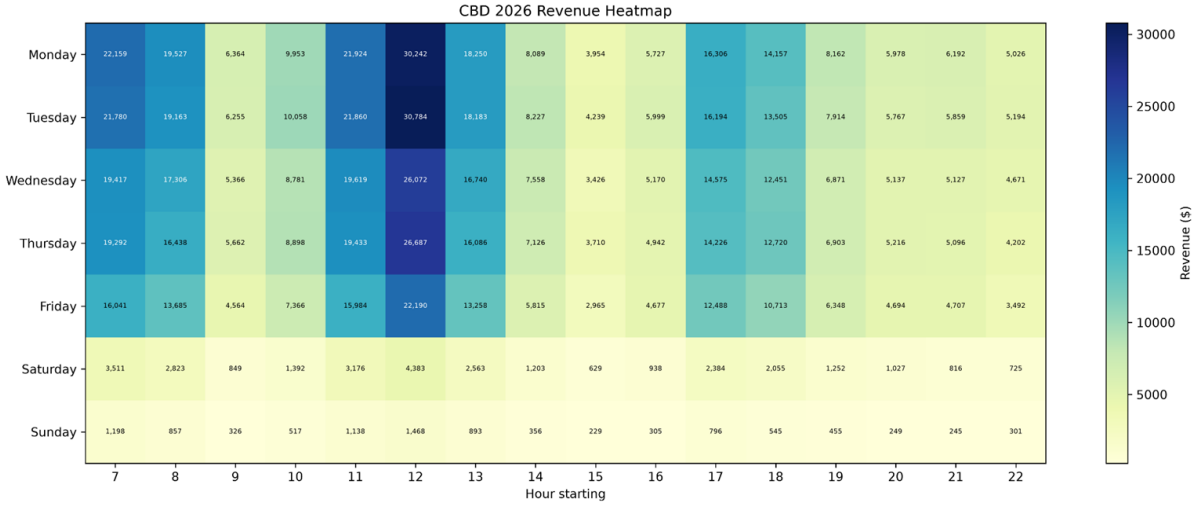


4.6. Cross-Outlet Performance Benchmark 2026

KPI	CBD	Orchard	TPY
Sales	946,504	1,508,913	1,494,418
Profit	57,047	223,790	263,219
Profit Margin	6.03%	14.83%	17.61%
Orders	54,444	55,136	50,355
Average Order Value	$17.38	$27.37	$29.68
Items / Order	2.83	3.65	4.03
Direct Contribution / Order	$10.08	$14.70	$17.02
Modelled Fulfilment Utilisation	40.85%	61.57%	58.48%
Weekend Revenue Share	4.18%	51.55%	46.78%

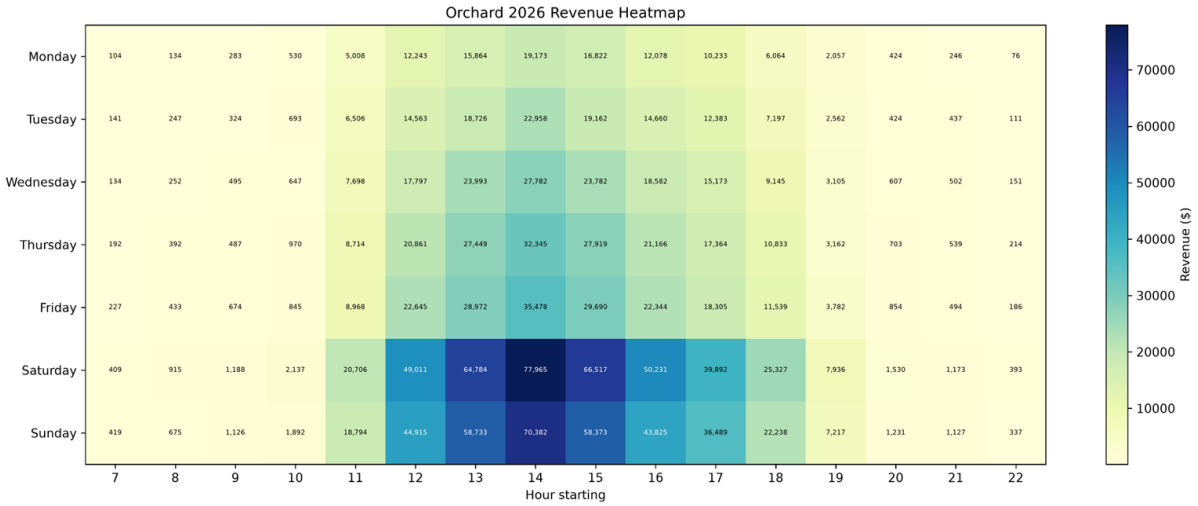
5. Appendix 5: Demand Pattern & Operating Analysis

5.1. CBD Demand Heatmap



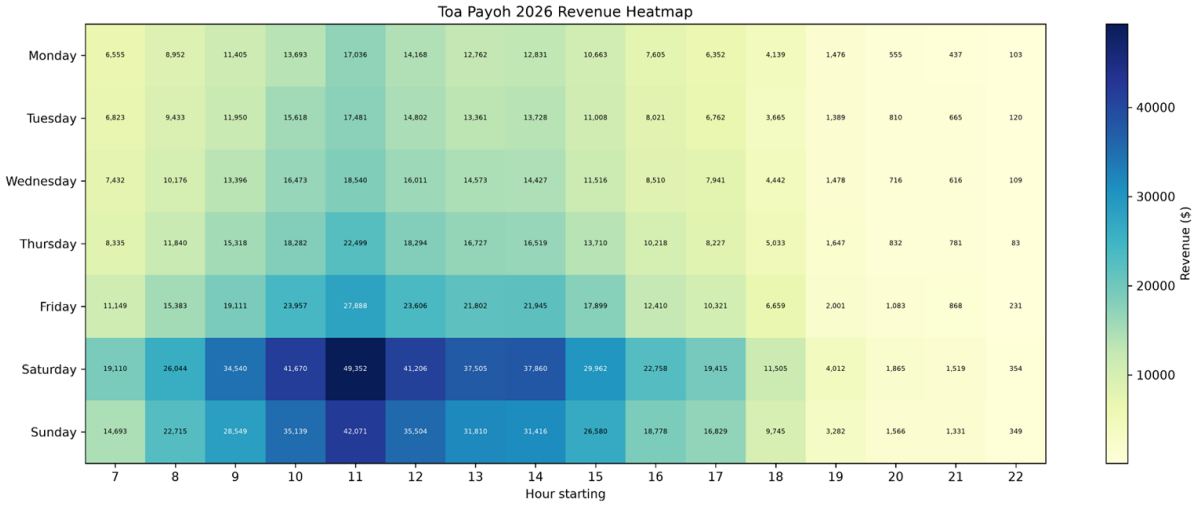
CBD demand is strongly concentrated on weekdays and around office-hour periods, while weekend demand is minimal.

5.2. Orchard Demand Heatmap



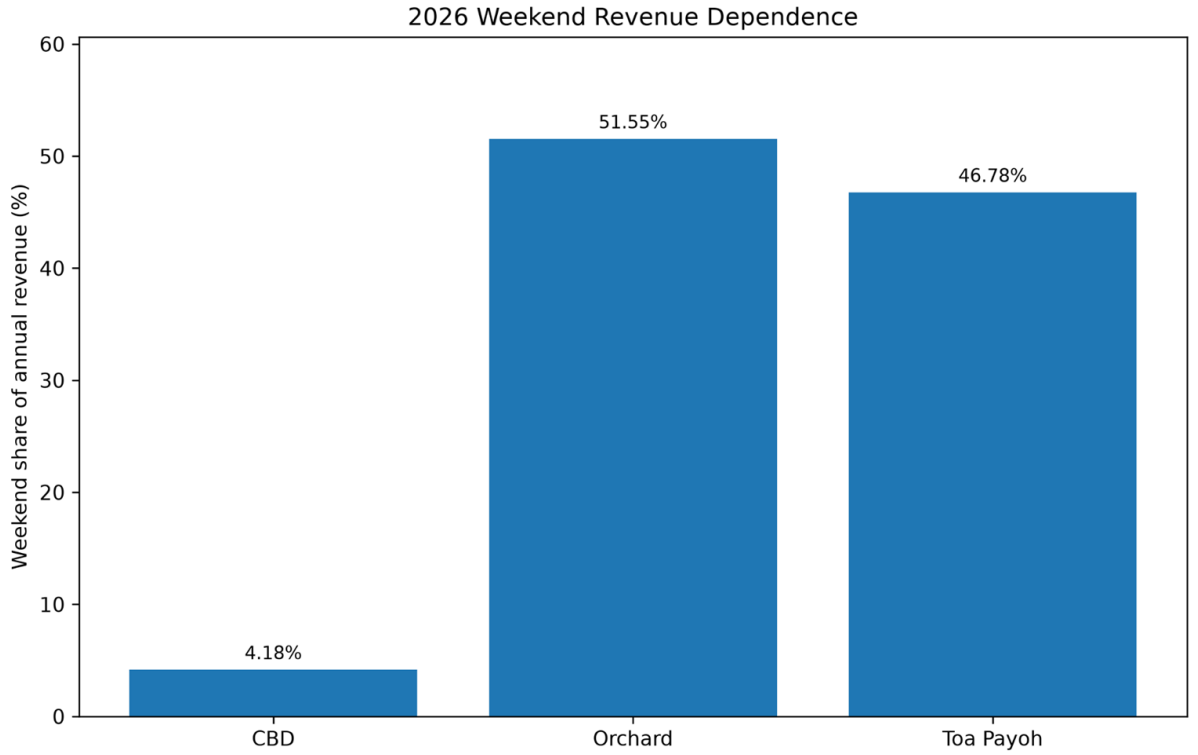
Orchard demand is concentrated around midday/afternoon and is particularly strong on weekends.

5.3. Toa Payoh Demand Heatmap



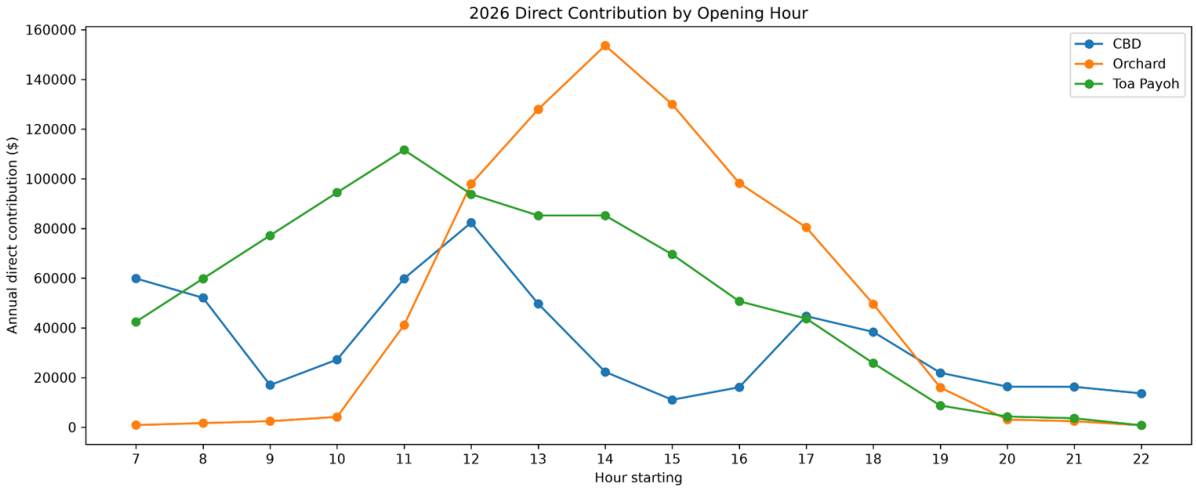
TPY exhibits broader daytime demand and relatively strong weekend activity. A uniform 7 am – 11 pm schedule is unlikely to work for all locations.

5.4. Weekend Revenue Dependence



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5.5. Hourly Contribution



	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
CBD	$59,892	$52,084	$16,970	$27,225	$59,836	$82,354	$49,724	$22,290	$11,057	$16,094	$44,721	$38,361	$21,931	$16,338	$16,279	$13,615
Orchard	$882	$1,665	$2,436	$4,167	$41,135	$97,942	$127,927	$153,642	$129,978	$98,273	$80,440	$49,560	$15,974	$3,095	$2,425	$778
Toa Payoh	$42,468	$59,817	$77,185	$94,460	$111,587	$93,807	$85,215	$85,220	$69,557	$50,702	$43,728	$25,786	$8,752	$4,314	$3,616	$775

6. Recommendation Economics & Sensitivity Analysis

6.1. Orchard Pricing Sensitivity: Incremental Annual Contribution

Price Increase	0% Volume Loss	-2% Volume Loss	-5% Volume Loss	-10% Volume Loss
+$0.25	50,248.50	33,037.21	7,220.27	-35,807.95
+$0.50	100,497.00	82,280.74	54,956.35	9,415.70
+$0.75	150,745.50	131,524.27	102,692.42	54,639.35
+$1.00	200,994.00	180,767.80	150,428.50	99,863.00

Positive values indicate scenarios in which the price increase produces greater contribution despite assumed volume loss.

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6.2. CBD Marketing Break-Even

Measure	Result
2025 Marketing Spend	$12,000
2026 Marketing Spend	$28,800
Additional Annual Marketing Spend	16,800
2025 Orders	54,869
2026 Orders	54,444
Actual Order Change	-425
Direct Contribution/2026 Order	$10.08
Break-Even Incremental Orders	1,667 orders
Assumed Consumables Pct of Sales	2.63%
Contribution after Estimated Consumables/Order	$9.62
Break-even Incremental Orders after Estimated Consumables	1,746 orders

The additional marketing expenditure requires a significant number of incremental orders merely to recover its cost, supporting stronger contribution-based campaign accountability.

6.3. Opening-Hour Sensitivity

Outlet	Scenario	Revenue at Risk	Contribution at Risk	Utility Saving	Net Before Labour - Consumables Not Assumed Variable	Net Before Labour - Consumables Assumed Variable
CBD	Close on weekends	39,603.00	22,950.84	12,000	-10,950.84	-9,908.03
ORD	11 am - 8 pm pilot	28,724.20	15,447.01	28,350	12,902.99	13,811.52
TPY	Closes after 8 pm	14,992.40	8,705.79	12,150	3,444.21	3,935.51

Note: Consumables are not assumed to be fully avoidable because their cost behaviour is not specified in the case. A separate sensitivity analysis assumes consumables vary

proportionately with sales. Sensitivity analysis, not a forecast. Labour savings, customer migration to alternative hours and behavioural responses are not fully observable from the case data.